

Examiner reports

Q1.

The responses to the modelling in parts (a) and (b) were very varied. Only a small number of candidates gained all three marks, but most gained at least one mark. Some candidates made statements that contradicted the information in the stem of the question, such as “the resistance force will be constant”. Part (c) tended to be done very well or very badly. There was some use of constant acceleration equations, but this was not very common. The biggest issue was the omission of a negative sign in the formation of the differential equation, which then continued throughout the solution. This error was quite common. Other problems were caused by incorrect algebraic manipulation, particularly with logarithms, or integration.

Q2.

Part (a) of this question was often done well. Many candidates seemed to take the question in their stride, but others made limited progress or did not know how to approach the question. There were some who lost accuracy marks. The four main causes of this were;

- Omission of the negative sign in the differential equation,
- Algebraic errors,
- Incorrect integration,
- Not finding the value of the constant of integration.

Part (b) was done very well, with many candidates gaining the available mark without doing part (a).

Q3.

Many candidates produced fully correct and succinct solutions here. Others barely scored any marks, as they seemed unaware of the variable nature of the acceleration and applied the equations of constant acceleration. A few started correctly, but made errors in separating variables prior to integration.

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Q5.

Part (a) was done very well, and many of the candidates seemed to have benefited from the printed answer. Part (b) however, produced a wide variety of responses. There was very little use of the constant acceleration equations in this question, which was encouraging. The main problems that the candidates encountered were with the integration. Errors such as omitting a negative sign or incorrectly splitting the $1/50 - v$ into two fractions were the most common errors.

Q6.

There were a number of good solutions to part (a). The main errors were due to

missing negative signs in the differential equation, the use of $v \frac{dv}{dx}$ instead of $\frac{dv}{dt}$ or algebraic errors in the working. Very few candidates tried to use the constant acceleration equations. Part (b) was answered correctly by the vast majority of

the candidates.



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